From X-ray to CT: A Comprehensive Survey of 2D-to-3D Reconstruction Methods From Classical Statistical Shape Models to Generative and Neural Implicit Frameworks

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Abstract

The reconstruction of three-dimensional (3D) volumetric anatomical structures from two-dimensional (2D) X-ray images is a fundamental challenge in medical imaging, with profound implications for clinical diagnosis, surgical planning, and patient-specific treatment. While Computed Tomography (CT) remains the gold standard for 3D anatomical visualization, its high radiation dose, cost, and limited accessibility have driven decades of research toward deriving volumetric information from low-dose, widely available X-ray projections. This survey provides a comprehensive and chronologically organized review of 2D-to-3D reconstruction methods spanning from classical statistical shape models and geometric registration techniques through early convolutional neural networks (CNNs), to the modern era of generative adversarial networks (GANs), diffusion models, and neural radiance fields (NeRF). Building upon the foundational taxonomy established by Maken and Gupta (2023), which systematically reviewed methods up to approximately 2020, this paper extends the timeline by critically analyzing the subsequent wave of data-driven deep generative frameworks. We examine landmark GAN-based methods including CVAE-GAN, X2CT-GAN, TRCT-GAN, DP-GAN+B, AP2CT-GAN, and PTX2CT-GAN, highlighting architectural innovations and clinical targets. We document the transition toward score-based diffusion models such as DX2CT and DVG-Diffusion, and review neural radiance field approaches including MedNeRF and X2BR. For each category, we analyze architectural design choices, datasets, evaluation metrics, anatomical scope, and clinical implications. Finally, we identify persistent challenges and outline promising research directions.

Keywords: X-ray to CT reconstruction; 2D-to-3D reconstruction; generative adversarial networks; diffusion models; neural radiance fields; medical image synthesis; deep learning; statistical shape models; volumetric reconstruction

1 Introduction

The human body is an inherently three-dimensional anatomical structure. Accurate visualization of internal anatomy is indispensable for modern clinical practice, encompassing disease diagnosis, preoperative planning, intraoperative guidance, and post-treatment follow-up [1]. Computed Tomography (CT) acquires hundreds of X-ray projections from different angles and reconstructs a volumetric image using filtered back-projection (FBP) or iterative reconstruction. The resulting volumetric data provides unambiguous information about internal structures, tissue densities, and spatial relationships.

However, CT imaging carries significant drawbacks. The radiation dose associated with a single CT examination can be 100 to 500 times that of a conventional X-ray, raising substantial concerns about cumulative radiation exposure, particularly for pediatric patients, pregnant women, and those requiring repeated imaging. CT scanners are also expensive, bulky, and unavailable in many resource-limited healthcare settings [31].

Plain X-ray radiography is the most widely deployed medical imaging modality worldwide. X-ray machines are inexpensive, portable, deliver orders-of-magnitude lower radiation doses, and produce images within seconds. Their fundamental limitation, however, is that an X-ray image is a two-dimensional projection of a three-dimensional structure. Overlapping anatomy, loss of depth information, and magnification distortion all limit its diagnostic utility in complex clinical scenarios [1].

This central tension—between the safety and accessibility of X-rays and the diagnostic richness of CT—has motivated decades of research into methods for reconstructing 3D volumetric information from 2D X-ray images. The field has evolved through four distinct technological eras, as illustrated in Fig. 1:

1. **Classical era (1990–2010):** Statistical shape models (SSMs), parametric anatomical models, point-based stereo correspondence, contour-based template deformation, and intensity-based registration [1].
2. **Early CNN era (2016–2020):** Convolutional neural networks that learn complex 2D-to-3D mappings directly from paired training data [6, 7, 8].
3. **GAN era (2019–2023):** Generative adversarial networks producing sharper, more realistic volumetric outputs [17, 21, 22].
4. **Diffusion / NeRF era (2022–present):** Score-based diffusion models and neural radiance fields enabling high-fidelity sparse-view reconstruction [24, 27, 28, 29].

The specific contributions of this paper are: (1) a unified recapitulation of classical and early-CNN methods estab-

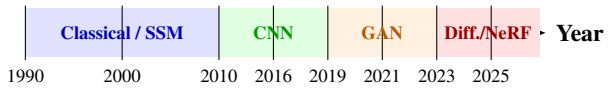

Figure 1. Chronological evolution of 2D X-ray to 3D CT reconstruction methods across four distinct eras.

lishing a common chronological baseline; (2) a structured review of GAN-based methods with detailed architecture diagrams; (3) analysis of the transition from GANs to diffusion models; (4) a review of neural radiance field approaches; and (5) a cross-cutting comparative analysis and discussion of open challenges.

The remainder of this paper is organized as follows. Section 2 recapitulates past work. Section 3 covers GAN-based methods. Section 4 discusses diffusion-based approaches. Section 5 reviews neural radiance field methods. Section 6 presents complementary CNN approaches. Section 7 provides comparative analysis. Section 8 discusses challenges and future directions. Section 9 concludes the paper.

2 Recapitulation of Past Work: From Statistical Models to Early CNNs

Before examining the modern generative era, it is essential to situate it within the trajectory of earlier methods. The foundational review by Maken and Gupta [1] provides the most comprehensive taxonomy of pre-2020 approaches across six categories: intensity-based methods, statistical shape models, articulated anatomical models, free-form and hierarchical deformation, convolutional neural networks, and hybrid methods. Figure 2 illustrates the general pipeline for 3D reconstruction from X-ray images.

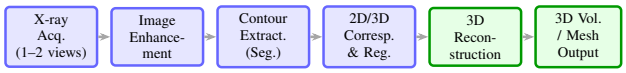

Figure 2. General pipeline for 3D anatomical reconstruction from 2D X-ray images applicable across all method categories [1].

2.1 Statistical Shape Models (SSMs) and Parametric Approaches

Statistical Shape Models, pioneered by Cootes et al. [5] through Point Distribution Models (PDMs), represent anatomical shape as a mean shape plus a weighted combination of principal components of shape variation:

$$\mathbf{S}' = M\bar{S} + M\sum_{j=1}^{s} f(\lambda_j)\mathbf{P}_j, \quad f(\lambda_j) = a_j = \sqrt{\lambda_j} \quad (1)$$

where $\bar{S}$ is the population mean shape, $\mathbf{P}_j$ are eigenvectors, λ_j are eigenvalues from PCA, and a_j are weight vectors. The covariance matrix is:

$$\text{Cov} = \frac{1}{p-1}\sum_{i=1}^{p}(S_i - \bar{S})(S_i - \bar{S})^T \quad (2)$$

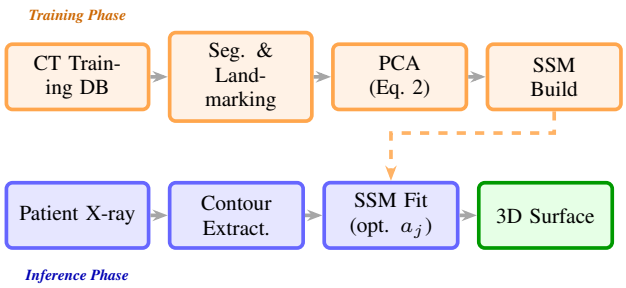

Figure 3. SSM-based reconstruction pipeline. An SSM is built offline from CT training data via PCA. At inference time, SSM parameters are optimized to match X-ray contours, yielding a patient-specific 3D surface model [9, 10].

Figure 3 illustrates the SSM-based reconstruction pipeline.

Lamecker et al. [9] demonstrated SSM-based reconstruction of the pelvic bone from synthetically generated X-rays, achieving symmetric mean errors of approximately 1.3–2.6 mm. Zheng [10] extended this to single-view AP X-ray reconstruction of the pelvis using a hybrid 2D/3D deformable registration strategy. Zhu and Li [11] applied enhanced SSM with 2D/3D registration to the distal femur, reporting an overall average absolute error of 0.90 mm. Parametric model-based reconstruction (Humbert et al. [12] for spine; Chaibi et al. [13] for lower limb) adjusts anatomically meaningful parameters to fit radiographic projections, achieving reconstruction times of 2–10 minutes.

2.2 Template Deformation Methods

Template deformation approaches deform a canonical 3D mesh model until its projected silhouette matches X-ray contours. Free-form deformation (FFD) was demonstrated by Koh et al. [14] for femur reconstruction. Hierarchical FFD by Gunay et al. [15] applies multi-resolution deformation for surgical osteotomy planning. Karade and Ravi [16] introduced Laplacian Surface Deformation (LSD) with Self-Organizing Maps (SOM) for 2D/3D correspondence, achieving mean point-to-surface distance of 1.2 mm in under 49 seconds.

2.3 Early CNN-Based Methods

The transition to data-driven approaches began with CNNs. Dixit et al. [7] proposed a feed-forward CNN to reconstruct 3D femur models from 30 X-ray samples. Kim et al. [8] combined CNN-based feature detection (YOLO v3 and Faster R-CNN) with SSM fitting for leg bone reconstruction, achieving 187-second computation time. The landmark contribution from Kasten et al. [6] introduced an end-to-end CNN for 3D knee bone reconstruction from biplanar X-rays using dimensional augmentation that back-projects epipolar line pairs into a two-channeled epipolar plane, achieving a Dice coefficient of 0.906 with 0.5-second inference—a step change in reconstruction speed.

3 GAN-Based Methods for X-ray to CT Reconstruction

Generative Adversarial Networks (GANs) [2] consist of a generator G trained to synthesize realistic outputs and a discriminator D trained to distinguish real from generated samples. In medical image synthesis, adversarial training produces volumetric CT outputs with sharper detail and more realistic texture than pure regression-based methods. Figure 4 illustrates the general GAN architecture for X-ray to CT synthesis.

3.1 Foundational Architecture: X2CT-GAN

X2CT-GAN established the canonical biplanar X-ray input paradigm for GAN-based volumetric synthesis. Taking orthogonal anteroposterior (AP) and lateral (LAT) chest X-ray images as input, it generates a full 3D chest CT volume via two separate 2D-to-3D encoding branches followed by feature fusion and a 3D decoder. The discriminator operates on both 2D projections of the generated volume and the 3D volume itself, exploiting complementary depth information from perpendicular views.

3.2 CVAE-GAN: Probabilistic 3D CT Reconstruction

Jiang et al. [18] proposed a Conditional Variational Autoencoder combined with a GAN discriminator (CVAE-GAN) for 3D CT reconstruction from a single X-ray, specifically targeting respiratory motion management in radiotherapy. The encoder maps a 2D X-ray projection into a latent distribution (μ, σ^2), from which a latent vector $\mathbf{z} \sim \mathcal{N}(\mu, \sigma^2)$ is sampled and decoded into a 3D CT volume. The training objective is:

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \beta\, D_{\text{KL}}\big(\mathcal{N}(\mu, \sigma^2) \,\|\, \mathcal{N}(0, I)\big) + \mathcal{L}_{\text{adv}} \quad (3)$$

By sampling from a learned latent distribution, the CVAE component enables representing the inherent ambiguity in reconstructing 3D anatomy from a single 2D projection, providing a clinically valuable implicit uncertainty estimate.

3.3 TRCT-GAN: Transformer-Enhanced Biplanar Reconstruction

Wang et al. [21] proposed TRCT-GAN, which enhances biplanar X-ray feature representation through transformer modules. While convolutional operations capture local spatial patterns efficiently, they struggle to model long-range dependencies such as the relationship between the left and right lung fields. TRCT-GAN incorporates a Transformer module within the generator to capture global contextual relationships, and a dynamic attention module to exploit both 2D and 3D feature maps. Figure 5 illustrates the dynamic attention mechanism.

3.4 DP-GAN+B: Lightweight GAN with Depthwise Separable Convolutions

Xing et al. [22] introduced DP-GAN+B to address the computational efficiency bottleneck limiting clinical deployment. DP-GAN+B replaces standard convolutions with depthwise separable convolutions:

$$\text{Params}_{\text{DW-Sep}} = D_K^2 C_{\text{in}} + C_{\text{in}} C_{\text{out}} \ll D_K^2 C_{\text{in}} C_{\text{out}} = \text{Params}_{\text{Std}} \quad (4)$$

achieving a parameter reduction of 31.924 million (44.17%) compared to X2CT-GAN baseline, while also introducing vector loss and fusion loss components.

3.5 AP2CT-GAN: Single AP View Reconstruction for ICU Settings

Tang et al. [25] proposed AP2CT-GAN for the ICU clinical scenario where only a single anteroposterior (AP) chest X-ray can practically be acquired. Two key modules are introduced: the Feature Enhancement Connection (FEC) module amplifying anatomically informative regions, and the Feature Dimension Converter (FDC) module bridging 2D and 3D domains. A Dual-Consistency Loss enforces structural and textural consistency:

$$\mathcal{L}_{\text{DC}} = \alpha\, \mathcal{L}_{\text{struct}} + (1 - \alpha)\, \mathcal{L}_{\text{texture}} \quad (5)$$

3.6 PTX2CT-GAN: Disease-Specific GAN for Pneumothorax

Wang et al. [26] presented PTX2CT-GAN targeting reconstruction of 3D chest CT from a single X-ray in the context of pneumothorax (PTX). The Reconstruction Feature Enhancement Module (RFEM) improves feature extraction specifically for pathological pneumothorax features. Trained and validated on the MY-PTX-998 dataset (998 pneumothorax chest X-rays with paired CT), PTX2CT-GAN demonstrates the value of disease-specific architectural design.

Table 1 summarizes all GAN-based methods reviewed.

4 Diffusion Models for X-ray to CT Reconstruction

Diffusion models (DDPMs) [3] operate by learning to reverse a gradual noising process. During training, Gaussian noise is progressively added to clean data over T timesteps:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}\big(\mathbf{x}_t; \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t \mathbf{I}\big) \quad (6)$$

The model learns to reverse this process, conditioned on X-ray images $\mathbf{c}$:

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{c}) = \mathcal{N}\big(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t, \mathbf{c}), \Sigma_\theta\big) \quad (7)$$

Diffusion models offer several advantages over GANs: training is stable (no adversarial game), mode coverage is better, and iterative refinement allows progressive improvement of reconstruction quality. Figure 6 illustrates the diffusion-based reconstruction framework.

4.1 DX2CT: Positional-Aware Diffusion for Bi/Mono-Planar Reconstruction

Jeong et al. [24] proposed DX2CT, a conditional diffusion model for 3D CT reconstruction from either biplanar or monoplanar X-ray images. DX2CT resolves the positional

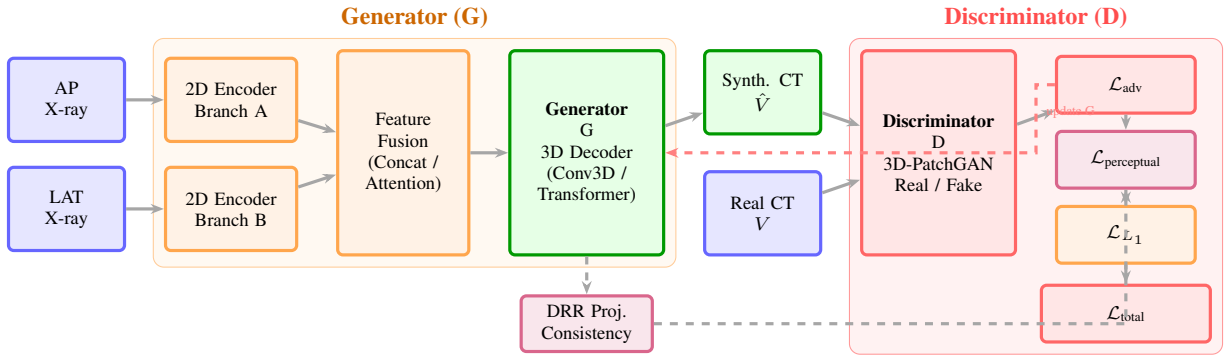

Figure 4. General GAN-based architecture for X-ray to CT reconstruction. Biplanar AP and LAT X-rays are independently encoded, fused, and decoded by a 3D generator. The discriminator distinguishes real from generated volumes. Multiple loss terms ($\mathcal{L}_{\text{adv}}$, $\mathcal{L}_{\text{perceptual}}$, $\mathcal{L}_{L_1}$) jointly optimize the generator [17, 21].

Table 1. Comparison of GAN-Based X-ray to CT Reconstruction Methods

Method	Year	Input	Region	Key Innovation	Dataset	Performance
CVAE-GAN [18]	2021	Single X-ray	Lung	Probabilistic VAE + GAN; uncertainty estimation	2 patient CTs	Feasibility; qualitative assessment
X-CTRSNet [19]	2022	Biplanar AP+LAT	C-Vertebra	Joint recon. + segmentation (RSC loss)	Institutional CT	PSNR=24.58 dB, SSIM=0.749, Dice=80.44%
XctNet [20]	2022	Single X-ray	General	Self-attn. + multi-scale + 3D branch gen.	Public dataset	PSNR=29.28 dB, SSIM=0.8681
TRCT-GAN [21]	2023	Biplanar	Chest CT	Transformer + dynamic cross-domain attention	ChestX-ray14/LIDC	Improved SSIM/PSNR vs X2CT-GAN
DP-GAN+B [22]	2024	Biplanar	Lung CT	Depthwise sep. convs.; 44.17% param reduction	Institutional CT	Comparable quality; -31.9M params
AP2CT-GAN [25]	2025	Single AP	Chest CT	FEC + FDC + dual-consistency loss	Institutional	Best single-view metrics vs baselines
PTX2CT-GAN [26]	2025	Single X-ray	Pneumothorax	RFEM for disease-specific features	MY-PTX-998	Superior PTX-specific quantitative metrics

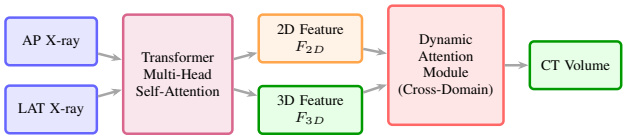

Figure 5. TRCT-GAN architecture [21]. A Transformer module captures long-range X-ray dependencies. A Dynamic Attention Module bridges 2D input features and 3D volumetric output features through cross-domain contextual association.

ambiguity of prior GAN-based methods through two key innovations. First, a novel transformer modulates 2D X-ray feature maps with 3D positional information of the CT volume via positional embeddings:

$$\mathbf{F}_{3D}^{pos}(i,j,k) = \text{CrossAttn}\big(\mathbf{E}_{pos}(i,j,k),\ \mathbf{F}_{2D}\big) \tag{8}$$

where $\mathbf{E}_{pos}(i,j,k)$ are 3D positional encodings of CT voxel (i,j,k) querying relevant 2D X-ray features $\mathbf{F}_{2D}$ in a geometrically principled manner. Second, these position-aware features condition the diffusion denoising network at each reverse step. Experiments on both biplanar and monoplanar X-ray benchmark datasets demonstrate that DX2CT outperforms several state-of-the-art methods.

4.2 DVG-Diffusion: Dual-View Guided Latent Diffusion

Xie et al. [27] proposed DVG-Diffusion, a dual-view guided diffusion model that directly addresses the ill-posed nature of mapping few-view 2D X-rays to 3D CT. The core insight is that a synthetic additional view can be generated and used alongside the real view to jointly guide reconstruction. DVG-Diffusion consists of three components: (1) a view-parameter-guided encoder extracting geometrically consistent features from real X-ray images; (2) a novel view synthesis module generating a synthetic X-ray from a different angle; and (3) a latent diffusion model conditioned on dual-view features, operating in the compressed latent space of a pre-trained VAE. Figure 7 illus-

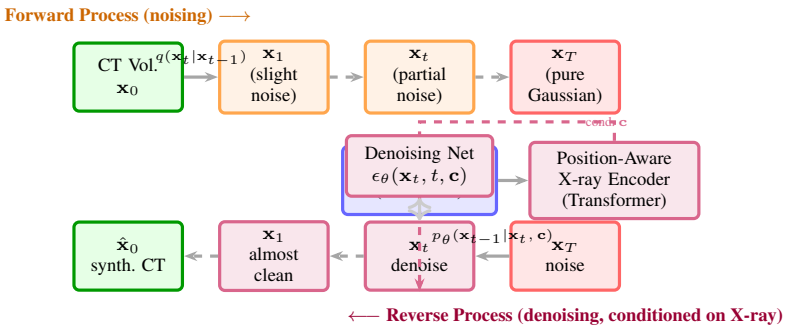

Figure 6. Diffusion model framework for X-ray to CT reconstruction. The forward process progressively adds Gaussian noise ($\mathbf{x}_0 \to \mathbf{x}_T$, Eq. 6). The conditioned reverse process denoises from pure Gaussian noise to produce synthetic CT $\hat{\mathbf{x}}_0$ [24, 27, 23].

trates this architecture.

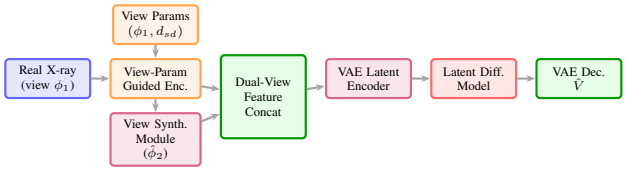

Figure 7. DVG-Diffusion architecture [27]. A view-parameter-guided encoder and view synthesis module generate dual-view features conditioning a Latent Diffusion Model (LDM) operating in VAE latent space.

4.3 High-Resolution 3D CT Synthesis Using Patch-Based Diffusion

Yoon et al. [23] proposed a 3D diffusion model for synthesizing high-resolution CT volumes from bidirectional X-ray images, targeting orthopedic applications (wrist fractures). To address computational cost, the 3D diffusion model is trained using patch-wise loss—operating on volumetric patches rather than the full CT volume, enabling single-GPU training. Patch coordinate information compensates for limited training examples by providing spatial context within the conditioning signal.

Table 2 summarizes diffusion-based methods.

5 Neural Radiance Fields for Medical 3D Reconstruction

Neural Radiance Fields (NeRF) [4] learn a continuous implicit function mapping a 3D position and viewing direction to density and radiance. In medical imaging, CT volumes can be interpreted as 3D density fields, and X-ray projections as rendered views. New views are synthesized via volumetric rendering:

$$C(\mathbf{r}) = \int_{t_n}^{t_f} T(t)\,\sigma(\mathbf{r}(t))\,\mathbf{c}(\mathbf{r}(t), \mathbf{d})\,dt, \quad T(t) = e^{-\int_{t_n}^{t} \sigma(\mathbf{r}(s))\,ds} \tag{9}$$

Figure 8 illustrates the NeRF framework adapted for medical imaging.

5.1 MedNeRF: Medical Neural Radiance Fields

Corona-Figueroa et al. [28] proposed MedNeRF, the first application of neural radiance fields to the problem of reconstructing 3D-aware CT projections from a single or few X-ray images. MedNeRF introduces a novel design that disentangles shape and volumetric depth, learning a continuous representation of CT scans by separately modelling surface geometry and internal density from 2D X-ray observations. Validated on chest and knee datasets, MedNeRF demonstrates that a single input X-ray is sufficient to synthesize new CT projections from arbitrary viewing angles. By reducing the number of X-ray projections required to infer 3D anatomy from hundreds (CT) to one, it has the potential to substantially reduce patient radiation exposure.

5.2 X2BR: Hybrid Neural Implicit Framework for Bone Reconstruction

Guven et al. [29] proposed X2BR, combining continuous volumetric reconstruction with template-guided non-rigid registration for high-fidelity 3D bone reconstruction from a single planar X-ray. X2BR comprises two components. The first, X2B, employs a ConvNeXt-based encoder to predict a 3D bone occupancy field using an implicit neural representation:

$$f_\theta : (\mathbf{x}, \mathbf{z}) \mapsto o \in [0, 1] \tag{10}$$

where $\mathbf{x} \in \mathbb{R}^3$ is a 3D query point and $\mathbf{z}$ is the X-ray latent code. X2B achieves IoU of 0.952 and Chamfer-L1 distance of 0.005, without relying on statistical shape models. The second component integrates anatomical priors via YOLOv9-based landmark detection and geodesic-based coherent point drift (CPD) registration, improving anatomical consistency (IoU: 0.875 for X2BR).

Table 3 summarizes NeRF and neural implicit methods.

6 Complementary CNN and Transformer Approaches

6.1 XctNet: Attention-Based Single-View Reconstruction

Tan et al. [20] proposed XctNet, introducing: (1) a self-attention module for adaptive feature optimization; (2) a

Table 2. Comparison of Diffusion-Based X-ray to CT Reconstruction Methods

Method	Year	Input	Region	Key Innovation	Dataset	Key Result
3D Diffusion [23]	2024	Bidirectional (2)	Wrist/Bone CT	Patch-wise loss; single-GPU; high-res synthesis	Institutional wrist	High-fidelity 3D CT; single GPU feasible
DX2CT [24]	2025	Mono or biplanar	Chest CT	3D position-aware transformer conditioning	LIDC-IDRI	SOTA on bi- and mono-planar benchmarks
DVG-Diffusion [27]	2026	Few-view	Chest/General	View synthesis + dual-view latent diffusion	ChestX-ray14/LIDC	Best fidelity-perceptual balance; SOTA IEEE TIP

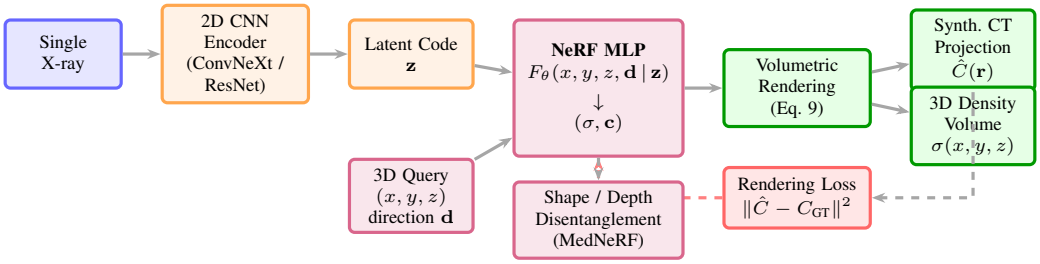

Figure 8. Neural Radiance Field (NeRF) architecture adapted for medical imaging. A 2D CNN encoder extracts latent code $\mathbf{z}$ from a single X-ray. The NeRF MLP maps 3D coordinates conditioned on $\mathbf{z}$ to density σ and radiance $\mathbf{c}$. Volumetric rendering synthesizes CT projections from arbitrary angles, and the 3D density volume constitutes the reconstructed CT [28, 29].

multi-scale feature fusion module integrating representations at multiple spatial resolutions; and (3) a 3D branch generation module producing volumetric outputs at different generation fields. XctNet achieved SSIM of 0.8681 and PSNR of 29.28 dB, significantly outperforming state-of-the-art methods at publication time.

6.2 X-CTRSNet: Joint Reconstruction and Segmentation

Ge et al. [19] proposed X-CTRSNet, the first framework for simultaneous 3D reconstruction and segmentation of cervical vertebrae CT directly from orthogonal 2D AP and lateral X-ray images. It couples SpaDRNet (reconstruction) and MulSISNet (segmentation) through a Reconstruction- Segmentation Consistency (RSC) learning strategy:

$$\mathcal{L}_{\mathrm{RSC}} = \mathcal{L}_{\mathrm{recon}} + \lambda\, \mathcal{L}_{\mathrm{seg}} + \mu\, \mathcal{L}_{\mathrm{consistency}} \tag{11}$$

achieving PSNR=24.58 dB, SSIM=0.749, and Dice coefficient of 80.44%.

6.3 AI-Driven CT Reconstruction: Clinical Perspectives

Pisuchpen et al. [30] provide a comprehensive review of AI in CT image reconstruction for abdominal imaging. Deep Learning-based Reconstruction (DLR) techniques leverage CNNs for high-fidelity denoising, enabling diagnostic-quality CT at substantially reduced radiation doses. These clinical advances establish the benchmark that synthetic CT from X-ray must meet for acceptance in radiology practice, and validate the architectural principles (CNNs for volumetric feature learning, perceptual losses, multi-scale processing) adopted in the synthesis literature.

7 Comparative Analysis Across Method Categories

7.1 Architectural Paradigm Evolution

Figure 9 illustrates the architectural progression from classical models to modern generative frameworks. The evolution follows a clear trajectory: from handcrafted geometric models requiring explicit anatomical knowledge, through discriminative CNN encoders, to generative frameworks modelling the full distribution of plausible 3D volumes.

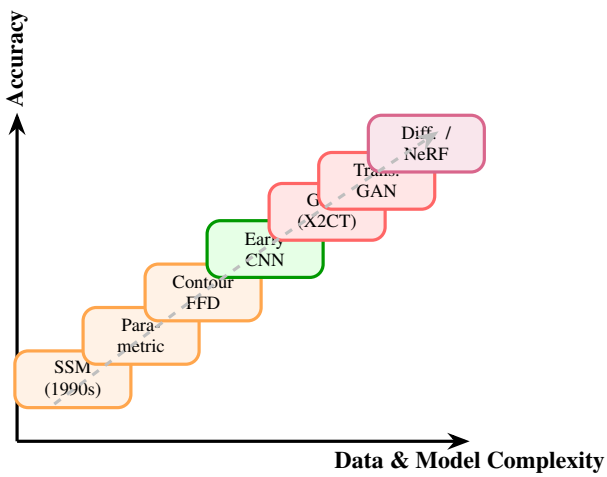

Figure 9. Qualitative illustration of architectural paradigm evolution in 2D X-ray to 3D CT reconstruction. Methods are plotted by data/model complexity (x-axis) and reconstruction quality (y-axis), showing a consistent upward trend.

Table 3. Comparison of Neural Radiance Field and Neural Implicit Methods

Method	Year	Input	Region	Representation	Dataset	Performance
MedNeRF [28]	2022	Single / few X-rays	Chest / Knee	NeRF + shape–depth disentanglement	Chest & knee CTs	High-fidelity CT projection synthesis; SOTA NeRF comparison
X2BR [29]	2026	Single planar X-ray	Bone (Rib, Vertebra)	ConvNeXt + neural occupancy + CPD registration	Institutional bone CT	X2B: IoU=0.952, Chamfer=0.005; X2BR: superior anatomical realism

7.2 Input Requirements and View Flexibility

Classical methods typically require two or more calibrated X-ray views with explicit correspondence annotations. Early CNNs and GANs use biplanar inputs. Recent methods—XctNet, AP2CT-GAN, PTX2CT-GAN, MedNeRF, and X2BR—push the frontier of single-view reconstruction, which is the most clinically practical scenario. DX2CT handles both mono- and biplanar inputs within a unified framework, providing flexibility for different clinical settings.

7.3 Anatomical Scope

Table 4 summarizes the anatomical coverage across methods. Classical methods were predominantly validated on skeletal structures (femur, tibia, pelvis, spine). Modern deep learning methods have substantially broadened the anatomical scope to include chest, lungs, cervical vertebrae, and pathological conditions (pneumothorax).

Table 4. Anatomical Scope of Representative Methods

Region	Classical	DL / Generative
Femur / Knee	Karade, Koh, Zhu, Kim	Kasten, DP-GAN+B
Pelvis	Lamecker, Zheng	—
Spine	Humbert, Cresson	X-CTRSNet
Chest / Lung	—	X2CT-GAN, TRCT-GAN, AP2CT-GAN, DVG-Diff
C-Vertebra	—	X-CTRSNet
Wrist / Bone	—	Yoon (Diff.), X2BR
Pneumothorax	—	PTX2CT-GAN
General	—	XctNet, MedNeRF

7.4 Cross-Era Quantitative Comparison

Table 5 provides a cross-era comparison of representative methods, highlighting the progressive improvement in quantitative metrics. A significant challenge in comparing methods across the literature is the heterogeneity of evaluation metrics: classical methods report geometric error metrics (mean point-to-surface distance, RMS error in mm) while modern methods report image quality metrics (SSIM, PSNR) on standardized datasets.

8 Challenges and Future Directions

Figure 10 summarizes the primary open challenges.

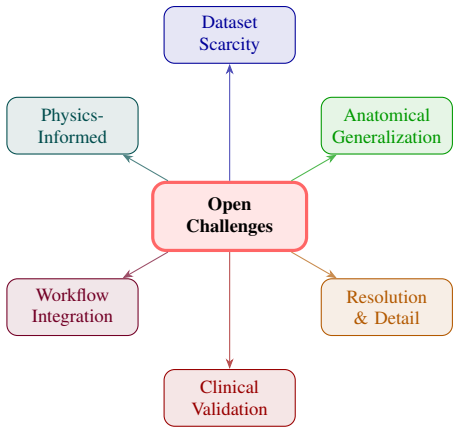

Figure 10. Six key open challenge areas in 2D X-ray to 3D CT reconstruction, each representing a significant research gap requiring dedicated investigation.

8.1 Dataset Scarcity and Ethical Constraints

The fundamental bottleneck for all data-driven methods is the availability of paired X-ray and CT data from the same patient. This is ethically and practically challenging—subjecting patients to additional radiation purely for research is generally not justified. Most studies rely on synthetic X-rays generated from CT data using digitally reconstructed radiograph (DRR) algorithms, but the domain gap between simulated and real X-ray appearance limits generalization. Future work should explore: (1) federated learning pooling data across institutions; (2) physics-based X-ray simulation with realistic noise and scatter; (3) self-supervised and semi-supervised learning paradigms; and (4) foundation model approaches pre-trained on large heterogeneous medical image datasets.

8.2 Anatomical Generalization

Most current methods are trained and validated on a single anatomical region and do not generalize to other body parts. The development of universal anatomical reconstruction models handling the full range of clinical X-ray examinations remains an open problem. Promising directions include multi-task learning with shared encoders and region-specific decoders, hierarchical models that first segment the region of interest, and foundation models trained on large-scale multi-anatomy datasets with anatomical

Table 5. Cross-Era Comparison of Key Methods Across All Four Eras

Method	Era	Type	Region	Input	Metric	Key Result
Karade et al. [16]	Classical	Contour+LSD	Femur	Biplanar	MP2S (mm)	1.2 mm; 49 sec
Kasten et al. [6]	Early CNN	CNN	Knee	Biplanar DRR	Dice / Chamfer	Dice=0.906; Chamfer=1.778 mm; 0.5 sec
X-CTRSNet [19]	CNN+GAN	Joint R+S	C-Vertebra	Biplanar	PSNR / SSIM / Dice	24.58 dB; 0.749; 80.44%
XctNet [20]	CNN	Attn-CNN	General	Single	PSNR / SSIM	29.28 dB; 0.8681
TRCT-GAN [21]	GAN	Trans.-GAN	Chest	Biplanar	SSIM / PSNR	Superior to X2CT-GAN baseline
MedNeRF [28]	NeRF	Neural Implicit	Chest/Knee	Single/few	SSIM / qual.	High-fidelity CT projection synthesis
DVG-Diffusion [27]	Diffusion	Latent Diff.	Chest	Few-view	SSIM / PSNR / FID	SOTA; best fidelity-perceptual balance
X2BR [29]	Neural Implicit	ConvNeXt+CPD	Bone	Single	IoU / Chamfer-L1	X2B: IoU=0.952; superior anatomical realism

conditioning.

8.3 Resolution and Fine Structural Detail

Current methods achieve promising results at moderate resolution (128^3 or 256^3 voxels) but struggle to reproduce fine anatomical detail—trabecular bone microstructure, small pulmonary nodules, delicate vascular structures. Cascade architectures generating a low-resolution coarse volume followed by super-resolution, and patch-based training (as in Yoon et al. [23]), represent promising approaches. Implicit neural representations offer theoretical resolution independence but face optimization challenges at fine scales.

8.4 Clinical Validation and Regulatory Pathway

The gap between research validation and clinical deployment is perhaps the most critical challenge. Current methods are evaluated on quantitative image quality metrics (SSIM, PSNR, Dice) that correlate imperfectly with clinical utility. Future work must develop: (1) task-specific evaluation protocols measuring radiologist performance in detecting specific pathologies; (2) prospective clinical studies validating diagnostic accuracy; (3) uncertainty quantification methods allowing clinicians to identify regions of low model confidence; and (4) regulatory frameworks for AI-synthesized medical images, which currently do not exist in most jurisdictions.

8.5 Integration with Clinical Workflow

For any reconstruction method to achieve clinical adoption, it must integrate with existing radiology workflows: inference times of seconds or minutes, integration with PACS and DICOM standards, robust performance across scanner vendors, and interpretable outputs. Lightweight architectures (DP-GAN+B's 44% parameter reduction) and efficient diffusion samplers represent steps toward practical deployment. Edge deployment on portable X-ray machines opens new possibilities for point-of-care 3D imaging in resource-limited settings.

8.6 Physics-Informed and Anatomically Constrained Models

Pure data-driven methods can produce anatomically implausible reconstructions—missing organs, distorted geometry, hallucinated structures—when extrapolating beyond the training distribution. Physics-informed neural networks (PINNs) can encode X-ray attenuation physics into the network. Biomechanical constraints (as in X2BR's template-guided registration) ensure physically plausible geometry. Integrating these constraints with diffusion models—whose iterative refinement is amenable to constraint injection at each denoising step—is a highly promising research direction.

9 Conclusion

This survey has traced the evolution of 2D X-ray to 3D CT reconstruction across more than three decades, from classical statistical shape models and geometric registration methods through the deep learning revolution and into the modern era of generative adversarial networks, diffusion models, and neural radiance fields. Building on the foundational taxonomy of Maken and Gupta [1], we have systematically reviewed eleven papers representing the state of the art as of 2025–2026, organized into four categories: GAN-based methods, diffusion-based methods, neural radiance field methods, and complementary CNN architectures.

Key insights from this survey are as follows. First, the field has undergone a clear paradigm shift from handcrafted geometric models to end-to-end learned generative frameworks, broadening anatomical scope, reducing input requirements from many calibrated views to a single X-ray, and dramatically improving reconstruction speed and quality. Second, GAN-based methods have established biplanar and single-view chest reconstruction as mature capabilities, while disease-specific adaptations (PTX2CT-GAN) demonstrate the value of targeted architectural design. Third, diffusion models offer stable training, better distributional coverage, and built-in uncer-

tainty quantification—DX2CT and DVG-Diffusion represent current state of the art. Fourth, neural radiance fields (MedNeRF, X2BR) introduce continuous implicit volumetric representations enabling extremely sparse-view reconstruction with significant radiation dose reduction potential. Fifth, hybrid methods combining neural implicit learning with anatomical priors (X2BR) provide the most clinically grounded reconstructions.

Despite these advances, critical challenges remain: dataset scarcity, anatomical generalization across body regions, clinical validation through task-specific evaluation and prospective studies, and regulatory frameworks for AI-synthesized medical images. Addressing these challenges will require collaboration across the deep learning research community, medical imaging, clinical radiology, regulatory agencies, and healthcare systems worldwide.

The vision motivating this field—providing 3D anatomical information at X-ray radiation doses and costs—is increasingly within reach. As methods reviewed in this survey continue to mature and clinical validation studies accumulate, X-ray to CT reconstruction has the potential to transform diagnostic imaging practice, bringing CT-level diagnostic power to settings that currently lack access while reducing radiation exposure for all patients.

References

[1] Maken P, Gupta A (2023) 2D-to-3D: A review for computational 3D image reconstruction from X-ray images. Arch Comput Methods Eng 30:85–114. https://doi.org/10.1007/s11831-022-09790-z

[2] Goodfellow I, Pouget-Abadie J, Mirza M et al (2014) Generative adversarial nets. In: Advances in Neural Information Processing Systems (NeurIPS) 27

[3] Ho J, Jain A, Abbeel P (2020) Denoising diffusion probabilistic models. NeurIPS 33:6840–6851

[4] Mildenhall B, Srinivasan PP, Tancik M et al (2020) NeRF: Representing scenes as neural radiance fields for view synthesis. ECCV 2020. Lect Notes Comput Sci 12346:405–421

[5] Cootes TF, Taylor CJ, Cooper DH, Graham J (1992) Training models of shape from sets of examples. Proc BMVC 1992:9–18

[6] Kasten Y, Doktofsky D, Kovler I (2020) End-to-end convolutional neural network for 3D reconstruction of knee bones from bi-planar X-ray images. In: Mach Learn Med Image Reconstr (MLMIR 2020). Springer, pp 123–133

[7] Dixit S, Pai VG, Rodrigues VC et al (2019) 3D reconstruction of 2D X-ray images. In: Proc 4th Int Conf CSITSS 2019, pp 1–5

[8] Kim H, Lee K, Lee D, Baek N (2019) 3D reconstruction of leg bones from X-ray images using CNN-based feature analysis. In: Proc ICTC 2019, pp 669–672

[9] Lamecker H, Wenckebach T, Hege H-C (2006) Atlas-based 3D-shape reconstruction from X-ray images. In: Proc ICPR 2006. IEEE, pp 371–374

[10] Zheng G (2010) Statistical shape model-based reconstruction of a scaled, patient-specific surface model of the pelvis from a single standard AP X-ray radiograph. Med Phys 37:1424–1439

[11] Zhu Z, Li G (2011) Construction of 3D human distal femoral surface models using a 3D statistical deformable model. J Biomech 44:2362–2368

[12] Humbert L, De Guise JA, Aubert B, Godbout B, Skalli W (2009) 3D reconstruction of the spine from biplanar X-rays using parametric models. Med Eng Phys 31:681–687

[13] Chaibi Y, Cresson T, Benjamin A et al (2012) Fast 3D reconstruction of the lower limb using a parametric model from biplanar X-rays. Comput Methods Biomech Biomed Eng 15:457–466

[14] Koh K, Kim YH, Kim K, Park WM (2011) Reconstruction of patient-specific femurs using X-ray and sparse CT images. Comput Biol Med 41:421–426

[15] Gunay M, Shim MB, Shimada K (2007) Cost- and time-effective three-dimensional bone-shape reconstruction from X-ray images. Int J Med Robot. https://doi.org/10.1002/rcs.162

[16] Karade V, Ravi B (2015) 3D femur model reconstruction from biplane X-ray images: a novel method based on Laplacian surface deformation. Int J Comput Assist Radiol Surg 10:473–485

[17] Ying X, Guo H, Ma K et al (2019) X2CT-GAN: Reconstructing CT from biplanar X-rays with generative adversarial networks. In: Proc CVPR 2019, pp 10619–10628

[18] Jiang L, Zhang M, Wei R et al (2021) Reconstruction of 3D CT from a single X-ray projection view using CVAE-GAN. In: Proc IEEE ICMIPE 2021. https://doi.org/10.1109/ICMIPE53131.2021.9698875

[19] Ge R, He Y, Xia C et al (2022) X-CTRSNet: 3D cervical vertebra CT reconstruction and segmentation directly from 2D X-ray images. Knowl Based Syst 236:107680. https://doi.org/10.1016/j.knosys.2021.107680

[20] Tan Z, Li J, Tao H, Li S, Hu Y (2022) Xct-Net: Reconstruction network of volumetric images from a single X-ray image. Comput Med Imaging Graph 98:102067. https://doi.org/10.1016/j.compmedimag.2022.102067

[21] Wang Y, Sun Z-L, Zeng Z, Lam K-M (2023) TRCT-GAN: CT reconstruction from biplane X-rays using transformer and generative adversarial networks. Digit Signal Process 140:104123. https://doi.org/10.1016/j.dsp.2023.104123

[22] Xing X, Li X, Wei C et al (2024) DP-GAN+B: A lightweight generative adversarial network based on depthwise separable convolutions for generating CT volumes. Comput Biol Med 174:108393. https://doi.org/10.1016/j.compbiomed.2024.108393

[23] Yoon S, Pratap JS, Liu W-C et al (2024) High-resolution 3D CT synthesis from bidirectional X-ray images using 3D diffusion model. In: Proc IEEE ISBI 2024. https://doi.org/10.1109/ISBI56570.2024.10635648

[24] Jeong YS, Yoo HB, Chun IY (2025) DX2CT: Diffusion model for 3D CT reconstruction from bi or mono-planar 2D X-ray(s). In: Proc IEEE ICASSP 2025. https://doi.org/10.1109/ICASSP49660.2025.10888986

[25] Tang T, Cai W, Yang B et al (2025) A novel medical image reconstruction framework for 3D chest CT based on a single 2D anteroposterior X-ray image. In: Proc IEEE SMC 2025. https://doi.org/10.1109/SMC58881.2025.11342980

[26] Wang Q, Tang T, Huang J et al (2025) PTX2CT-GAN: Reconstruction of chest CT from single X-ray radiography for pneumothorax. In: Proc IEEE BIBM 2025. https://doi.org/10.1109/BIBM66473.2025.11356273

[27] Xie X, Liu J, Fan H et al (2026) DVG-Diffusion: Dual-view-guided diffusion model for CT reconstruction from X-rays. IEEE Trans Image Process 35:1158–1172

[28] Corona-Figueroa A, Frawley J, Bond-Taylor S et al (2022) MedNeRF: Medical neural radiance fields for reconstructing 3D-aware CT-projections from a single X-ray. In: Proc IEEE EMBC 2022. https://doi.org/10.1109/EMBC48229.2022.9871757

[29] Guven G, Ugurdag F, Ates HF (2026) X2BR: High-fidelity 3D bone reconstruction from a planar X-ray image with hybrid neural implicit methods. Med Biol Eng Comput. https://doi.org/10.1007/s11517-026-03549-5

[30] Pisuchpen N, Srinivas Rao S, Noda Y et al (2026) Artificial intelligence (AI) and CT in abdominal imaging: image reconstruction and beyond. Abdom Radiol 51:424–445. https://doi.org/10.1007/s00261-025-05031-6

[31] Gupta A (2020) Challenges for computer aided diagnostics using X-ray and tomographic reconstruction images in craniofacial applications. Int J Comput Vis Robot 10:11. https://doi.org/10.1504/IJCVR.2020.10029170